

Lab 3 – Lab Reflection

SRN: PES1UG24AM221

System: Domain & SSL Certificate Expiry Alert System

1. What I Learned

In this lab, I learned how architectural styles affect the structure and behaviour of a software system. I compared Layered, Microservices, and Client-Server architectures and understood why the choice of architecture should be based on the requirements of the system. I also learned how UML component diagrams represent components, interfaces, dependencies, and interactions.

2. Architectural Decision

For the Domain & SSL Certificate Expiry Alert System, I selected **Microservices Architecture**. The system contains separate functions such as WHOIS auditing, SSL/TLS auditing, expiry detection, and alert management. Treating these functions as separate services provides modularity, scalability, and fault isolation.

3. Component Modelling Experience

I identified the major components of the system and modelled how they communicate through interfaces. The main components are the Web Application, Domain Monitoring Manager, WHOIS Audit Service, SSL/TLS Audit Service, Expiry Detection & Alert Service, and Audit Database. I also used interfaces such as Domain Management API, WHOIS Audit API, SSL/TLS Audit API, and Alert API to represent communication between components.

4. Challenges Faced

One challenge was keeping the component diagram clear while showing the required interfaces and data flows. Too many connections made the diagram crowded, so I learned to focus the component model on the most important architectural interactions and arrange the components logically.

5. Skills Developed

This lab improved my understanding of architectural pattern selection, UML component modelling, interface representation, dependency analysis, and architectural documentation. I also gained experience in presenting a system architecture in a clear and structured way.

6. Overall Reflection

Overall, this lab helped me understand that an architecture should not be selected only because it is popular, but because it addresses the specific functional, security, performance, scalability, and reliability needs of the system. The exercise gave me practical experience in converting system requirements into an architectural component model.